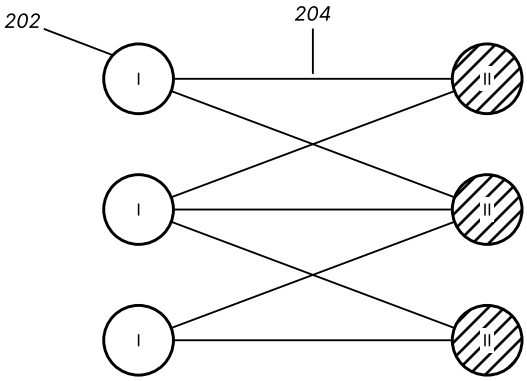
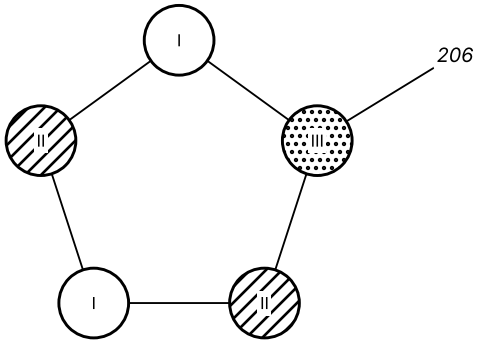
INTERACTION GRAPHS PARTITIONED INTO UPDATE CLASSES

(A) BIPARTITE — 2 CLASSES



every edge joins I to II

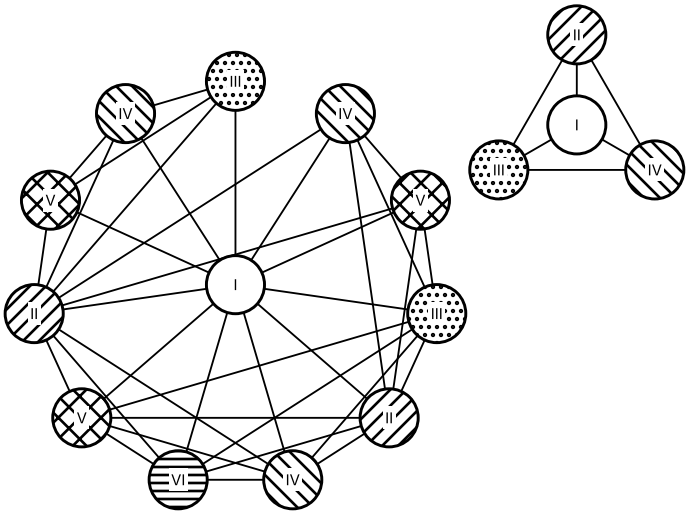
(B) ODD CYCLE — 3 CLASSES



a 5-cycle cannot be 2-coloured

(C) MEASURED NETWORK, PROPER COLOURING

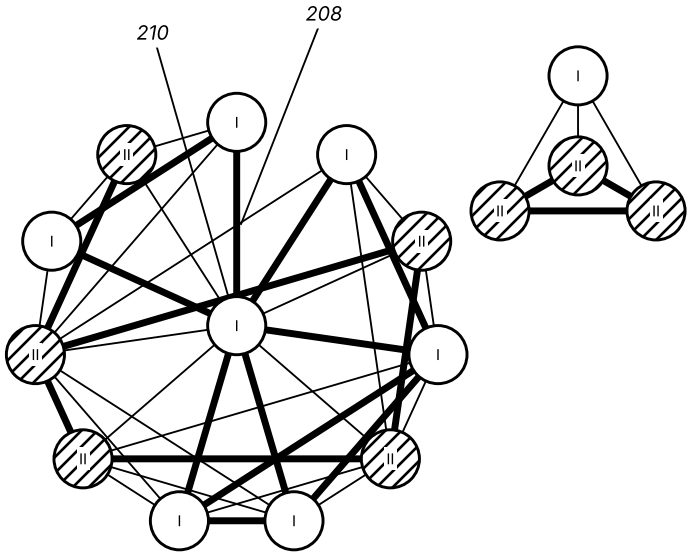
N = 16, 43 couplings, 6 classes



0 of 43 coupled pairs share a class

(D) SAME NETWORK, PARTITIONED BY INDEX PARITY

2 classes (even / odd) — NOT a proper colouring



19 of 43 coupled pairs (44.2%) share a class (bold, 208)

210 highest-degree node: 6 of its 11 neighbours in its own class

MEASURED — UPDATE CLASSES REQUIRED, SPARSIFIED AND RETRAINED NETWORKS (FIG. 7)

nodes N	16 [1]	256	1 024	4 096
classes K	6	4	5	6

[1] *the N = 16 network of (C)–(D) is dense (35.8%) and not of bounded degree; it is shown for the parity comparison.*

For interaction graphs of bounded degree the number of classes does not grow appreciably with N. The colouring need not be optimal.

FIG. 2